Simultaneous Speech and Eating Behavior Recognition Using Multitask Learning

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***Abstract*—The importance of "talking" and "eating" in preserving one's well-being has been underscored. Automatic recognition of daily conversation and eating behavior has promising implications for healthcare management and monitoring systems for the older population. In recent years, the performance of speech recognition models has improved dramatically. However, incorporating sounds produced during eating behaviors, particularly those associated with actions such as chewing and swallowing, into current speech recognition models is challenging. Moreover, the simultaneous recognition of speech and eating behaviors has not been accomplished with sufficient accuracy, despite their high relevance to speech. This study proposes a model for simultaneous speech and eating behavior recognition through multitask learning of two tasks: "speech recognition" and "eating behavior recognition." The model utilizes data collected using a biological sound collection device that makes direct contact with the skin. To validate the effectiveness of the proposed method, we conducted evaluation experiments on a publicly available dataset with respect to participants and corpus.**

Keywords—speech recognition, eating behavior recognition, multitask learning, wav2vec2.0

I. Introduction

In recent years, speech recognition using end-to-end architecture has made remarkable progress [1]. Utilizing this model, scholars have attempted to recognize eating behavior using biological sounds recorded by a skin-contact microphone [2]. Moreover, attempts have been made to use this skin-contact microphone for speech recognition [3]. Although speech and eating behavior are known to provide important information about the health of the older people, models that can process both simultaneously and efficiently have yet to be fully investigated. Using the SSL model (wav2vec2.0[4]) as a foundation, this study proposes a method for efficiently obtaining effective latent representations for discriminating various sounds and solving the automatic speech recognition (ASR) and eating behavior recognition (EBR) in an integrated manner through multitask learning. By treating both tasks as interconnected and sharing the neural network's intermediate representation, we aim to develop a comprehensive model that can accurately recognize speech and eating behavior in a meal scene in an integrated manner while enhancing the accuracy of each task.

II. Dataset

A. Skin-Contact Microphone

Fig. 1. shows images of a throat microphone and a below- ear microphone. These microphones are skin-contact microphones that directly record skin vibrations, thus

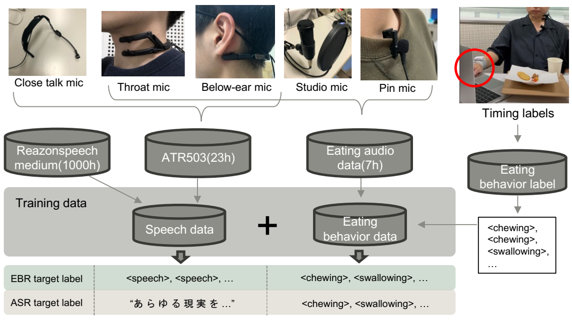

Fig. 1. Overview of training data

effectively capturing the sounds of eating behavior associated with chewing and swallowing. Two channels are placed at the upper and lower positions of the throat, whereas two channels are placed at the right and left positions below the ear.

B. Training Data

A summary of training data is shown in Fig. 1. The training data consist of speech and eating behavior data. For the speech data, we used Reazonspeech — medium [5], a corpus of about 1,000 h of Japanese speech, and approximately 23 h of ATR503 speech recorded by 30 participants in our laboratory. Skin-contact microphones have different acoustic characteristics than conventional microphones because they record skin vibrations directly. To eliminate this mismatch, we trained on 5 channel simultaneous recordings of ATR503 sentences with balanced phoneme frequencies from skin-contact and closed-talk microphones. We used approximately 7 h of eating behavior sounds (chewing: 32,793 times, swallowing: 1,426 times) from 59 participants collected in our laboratory. The food items were cabbage, crackers, and gum. 6 channels were recorded simultaneously using a skin-contact microphone, a studio microphone, and a pin microphone. During the recording, the participants performed button operations at the same time they chewed and swallowed, and the resulting weak behavioral labels served as the target event sequence. The training data were then labeled with one of two types of targets. EBR target labels were assigned to the labels corresponding to the <speech>, <chewing>, and <swallowing> events in the input data. In the case of speech data, ASR target labels were applied to the transcription results. Meanwhile, eating behavior data use the same EBR target labels.

C. Test data

we used a newspaper-read speech as the test data. The test speakers were 7 participants who did not participate in the training speakers (speech: 280 times, chewing: 4,150 times, swallowing: 150 times). The input speech consisted of three

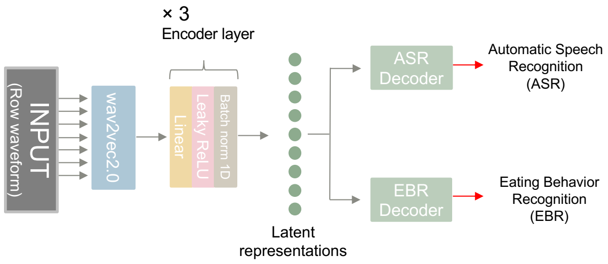

Fig. 2. Multitask learning model

channels of speech data from throat microphones (upper and lower) and a below-ear microphone (left). The data were not preintegrated between the microphones, and each channel received independent speech data for evaluation.

III. Proposed Method

Fig. 2. depicts the proposed multitask learning model for this study. Wav2vec2.0 [4] is a self-supervised learning model that uses raw waveform data to generate a latent representation known to perform well in downstream tasks, primarily multilingual speech recognition. The model employs a 1,024-dimensional wav2vec2.0 latent representation with three linear layers, and the resulting intermediate features are branched and analyzed using a decoder with different output dimensions. The output of the ASR shows the results of the speech recognition in the input speech. Meanwhile, the output of the EBR shows the recognition results of the event sequences <chew>, <swallow>, <speech>, and <blank> in the input speech. The loss function was based on the CTC loss, which is known to perform well in speech recognition tasks that use sequence data. Among other sequence data characteristics, eating behavior frequently includes a tendency to swallow after chewing. Therefore, we also used CTC loss for this task. This model has two outputs, but to capture the effect of multitask learning, each loss was weighted by the coefficient α and combined to form the loss (Equation (1)). In the equation, $ASR_{CTCloss}$ denotes the loss on the speech recognition side, whereas $EBR_{CTCloss}$ represents the loss on the EBR side.

$$Loss = \alpha \times ASR_{CTCloss} + (1 - \alpha) \times EBR_{CTCloss} \quad (1)$$

IV. Experiments

A. Experimental setup

We used wav2vec2-large-xlsr-53 as a pretraining model for wav2vec2.0, freezing only the parameters for the Feature Extractor. Four Nvidia V100 SXM2 16GB cards were used to train 40 epochs over approximately 50 h. The value of α was set to 0.75 for the first 20 epochs, and then to 0.5. To evaluate the performance of the event classification, we used the F1-score metric. We generated simulated data by combining test data, with some utterances containing eating behavior. The model's output was assessed at 5 msec intervals.

Furthermore, the character error rate (CER) was calculated to evaluate the accuracy of both the speech recognition and outputting event sequences, such as the order and counts of eating behaviors. In this evaluation, test data were not combined, and the model was tested on speech and eating behavior independently.

B. Experimental results

Table I shows the classification performance of speech and eating behavior using a skin-contact microphone. The results show that speech can be classified with high accuracy, but there is room for improvement in the accuracy of eating behavior classification. Table II shows the results of the CER evaluation of the accuracy of speech and event sequences of eating behavior. The CER values for all skin-contact microphones were in the 10% range. Regarding the CER of speech sounds, the results suggest that the model performs reasonably well in recognizing speech content, albeit a bit worse than existing models. In terms of eating behavior, swallowing was often misrecognized as chewing. While the CER for eating behavior was similar between the throat and below-ear microphones, the throat microphone showed better performance in accurately detecting swallowing events.

TABLE I. EVENT CLASSIFICATION PERFORMANCE (F1-SCORE)

Event	***Throat microphone (upper)***	***Throat microphone (lower)***	***Below-ear microphone (left)***
Speech	0.89	0.89	0.88
Eating behavior	0.68	0.51	0.72

TABLE II. SPEECH AND EATING BEHAVIOR RECOGNITION RESULTS (CER[%])

Event	***Throat microphone (upper)***	***Throat microphone (lower)***	***Below-ear microphone (left)***
Speech	11.34	12.97	12.06
Eating behavior	13.8	17.83	13.8

V. Conclusion

In this study, we constructed a model for recognizing simultaneous speech and eating behavior using multitask learning. We assessed the accuracy of the model on an open dataset, considering both the participants and the corpus. However, there is still room for enhancing the accuracy of each individual task. In the future, we will analyze data augmentation techniques for speech data, including chewing and swallowing sounds, and attempt to evaluate the model's performance using actual data from spontaneous speech and eating behavior.

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